

QRM2

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Week 1

1 Dataset

Find the dataset “movies5.tsv” on Canvas. Describe your data set: How many observations does it have? How many variables are there? How many subjects? What consists of a subject? [4 points]

```
# Load the dataset
movies <- read.delim("movies5.tsv")

nrow(movies)
## [1] 909
ncol(movies)
## [1] 19

length(unique(movies$index))
## [1] 909
```

Your Answer:

Number of observations: 909 observations (rows).

Number of variables: 19 variables (columns).

Number of subjects: 909 subjects (films).

The dataset contains 909 observations and 19 variables, including the identifier “index”. There are 909 subjects, each representing an individual movie. Each row contains one observation with information about a movie, such as its title, budget, revenue, and runtime. Since each movie appears exactly once in the dataset, the number of observations equals the number of subjects.

2 Types of variables

Which of the following types of variables are present in your data set? (i) nominal; (ii) ordinal; (iii) interval; (iv) ratio. If present, name one example of such a variable present in your data set. [4 points]

```
# Show the variables and their data types
str(movies)
## 'data.frame':    909 obs. of  19 variables:
## $ index          : int  4749 1744 960 1810 2889 1527 3520 2050 3755 1390 ...
## $ budget         : num  0.0 3.0e+07 5.0e+07 3.0e+07 1.2e+06 ...
## $ keywords       : chr  "love independent film best friend true love irish catholic" "alcohol one-night stand" ...
## $ original_language : chr  "en" "en" "en" "en" ...
## $ title          : chr  "The Brothers McMullen" "Knocked Up" "The Adventures of Sharkboy and Lavagirl" "The Pacifier" ...
## $ popularity      : num  1.58 22.53 12.31 38.17 3.58 ...
## $ release_date    : chr  "1995-08-09" "2007-06-01" "2005-06-10" "2004-02-25" ...
## $ revenue         : num  0.00 2.19e+08 6.94e+07 6.12e+08 0.00 ...
## $ runtime        : int  98 129 92 127 86 137 112 96 93 92 ...
## $ vote_average    : num  6.3 6.2 4.4 6.9 6.2 6.9 6.2 5.2 5.3 6.2 ...
## $ vote_count      : int  23 1214 269 864 50 113 578 559 26 1428 ...
## $ genre           : chr  "Drama" "Drama" "Adventure" "Drama" ...
## $ release_year     : int  1995 2007 2005 2004 2002 1995 2002 2015 2001 2009 ...
## $ release_month    : int  8 6 6 2 5 12 4 9 11 1 ...
## $ release_day      : int  9 1 10 25 24 29 11 3 16 22 ...
## $ first_actor      : chr  "Shari Albert" "Seth Rogen" "Taylor Lautner" "Jim Caviezel" ...
## $ first_actor_gender : chr  "female" "male" "female" "male" ...
```

```
## $ director_first_name: chr "Edward" "Judd" "Robert" "Mel" ...
## $ director_gender : chr "male" "male" "male" "male" ...

unique(movies$genre)
## [1] "Drama" "Adventure" "Comedy" "Thriller" "Documentary"
## [6] "Action" "Fiction" "Music" NA "Animation"
## [11] "Family" "Horror" "Romance" "Western" "Crime"

sort(unique(movies$vote_average))
## [1] 0.0 2.3 2.8 2.9 3.0 3.2 3.3 3.4 3.7 3.8 3.9 4.0 4.1 4.2 4.3 4.4 4.5
## [18] 4.6 4.7 4.8 4.9 5.0 5.1 5.2 5.3 5.4 5.5 5.6 5.7 5.8 5.9 6.0 6.1 6.2
## [35] 6.3 6.4 6.5 6.6 6.7 6.8 6.9 7.0 7.1 7.2 7.3 7.4 7.5 7.6 7.7 7.8 7.9
## [52] 8.0 8.3 9.5

head(movies$release_year)
## [1] 1995 2007 2005 2004 2002 1995
head(movies$runtime)
## [1] 98 129 92 127 86 137
```

Your Answer:

- (i) “Genre” is a nominal variable, since categories like Comedy, Drama, and Action don’t have any natural order.
- (ii) The dataset does not include an ordinal variable. We treat “vote_average” as an interval variable, which assumes equal distances between rating points.
- (iii) “Release_year” qualifies as an interval variable. The gap between two years always means the same thing—ten years is ten years—yet there’s no true zero. So, moving from 2000 to 2010 is just as much time as 2010 to 2020.
- (iv) “Runtime” is a ratio variable. Time has a true zero, and we can talk about its values in terms of multiples—a 120-minute movie is twice as long as a 60-minute one.

3 Movie profits and ratings

A movie studio wants to know which types of movies give maximal profit. Perform the following steps to provide the movie studio with an analysis which corresponds to their request:

- a. Create the variable profits as the revenue of a movie minus its budget. Report its mean, median, maximum, and minimum. [2 points]
- b. Which movie has the highest profits in your data set and how much are these profits? Which movie has the lowest and how much are its profits? If multiple movies have the exact same highest or lowest profits, give only one example. [2 points]
- c. Create a boxplot of the variable profits. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Give Q1, Q2, Q3, and Q4. What does this tell you about the nature of making money in the movies industry? [2 points]
- d. Add a new variable to your data set the log of profits. When creating this variable, what happens to movies for which profits is zero or negative? What then happens when you calculate the mean of log of profits? [2 points]
- e. For movies that have a profit of zero or less, replace log of profits with “NA”. What is now the mean of log of profits? Create a boxplot for log of profits, again with an appropriate title, x- and y-axis labels. How does it compare to the boxplot you made under c.)? [2 points]
- f. Create a scatterplot of with the runtime of movies on the x-axis and the average vote of movies on the y-axis. What do you conclude from the scatterplot? Are movies with a longer runtime considered worse or better by the audience, or does the audience not have a preference? Why do you think this is the case? [2 points]

For each step, you should provide first all the code you used to answer the question and then formulate an answer using full sentences.

Step a

```
# Create profits variable
movies$profits <- movies$revenue - movies$budget

mean(movies$profits)
## [1] 56284650

median(movies$profits)
## [1] 2942221

max(movies$profits)
## [1] 1082730962

min(movies$profits)
## [1] -98301101
```

Your Answer:

We calculated profit by subtracting the budget from the revenue. The mean profit is 56,284,649.70, and the median is 2,942,221. The highest profit is 1,082,730,962, and the lowest is −98,301,101.

Step b

```
# Movie with the highest profit and its profit
movies$title[which.max(movies$profits)]
## [1] "Minions"
max(movies$profits)
## [1] 1082730962

movies$title[which.min(movies$profits)]
## [1] "The 13th Warrior"
min(movies$profits)
## [1] -98301101
```

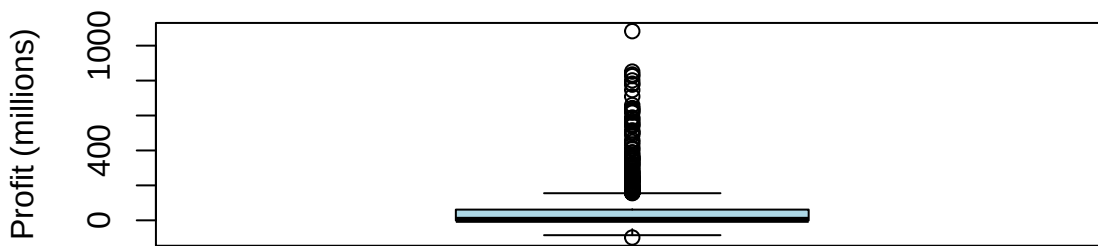
Your Answer:

“Minions” has the highest profit, earning 1,082,730,962 above its budget. “The 13th Warrior” has the lowest profit, with a loss of 98,301,101. This result was also found with max/min in step a.

Step c

```
# Boxplot of profits in millions
boxplot(movies$profits / 1000000,
        main = "Movie profits",
        xlab = "All movies",
        ylab = "Profit (millions)",
        col = "lightblue")
```

Movie profits



All movies

```
summary(movies$profits)
##      Min.      1st Qu.      Median      Mean      3rd Qu.      Max.
## -98301101 -1260859    2942221    56284650    61340801 1082730962
```

Your Answer:

Q1 is $-1,260,859$, Q2 is $2,942,221$, and Q3 is $61,340,801$. We interpret Q4 as the maximum, which is $1,082,730,962$. The boxplot shows that a few movies make much higher profits than the rest, which pulls the mean above the median. Based on the recorded revenue and budget, at least a quarter of the movies made a loss. This suggests that making movies can be risky: some earn huge profits, while many others earn much less or lose money.

Step d

```
# Create the natural logarithm of profits
movies$log_profits <- log(movies$profits)
## Warning in log(movies$profits): NaNs produced

# Count movies with zero or negative profits
sum(movies$profits == 0)
## [1] 162
sum(movies$profits < 0)
## [1] 261

mean(movies$log_profits)
## [1] NaN
```

Your Answer:

For movies with zero profit, taking the logarithm gives $-\text{Inf}$. For movies with negative profit, it gives NaN, and R shows a warning. There are 162 movies with zero profit and 261 with negative profit. Because the log-profit variable contains these values, it means the calculation of the mean becomes NaN. You would have to filter out these NaN and $-\text{Inf}$ to correctly calculate the mean of movies with more than 0 profit.

Step e

```
# Replace log profits with NA for non-positive profits
movies$log_profits[movies$profits <= 0] <- NA

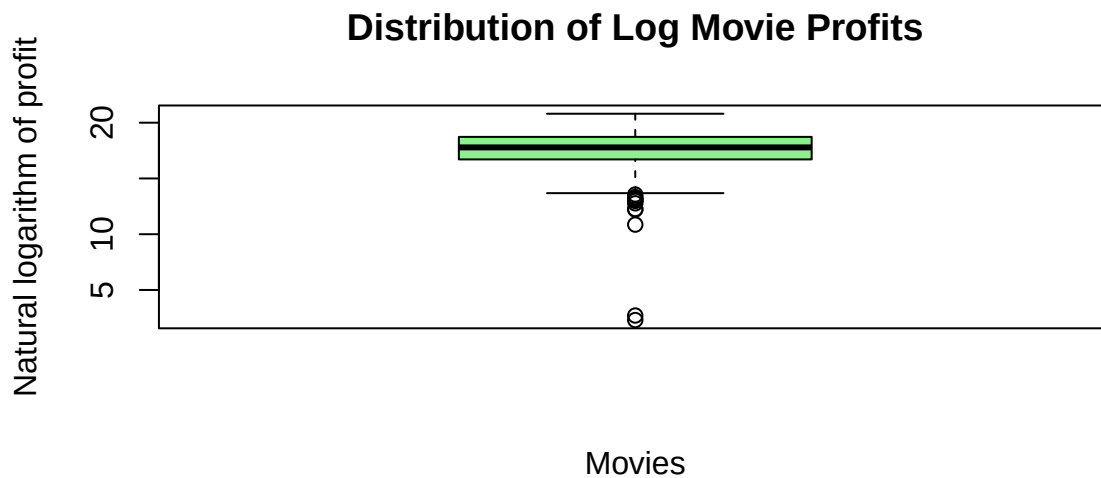
mean(movies$log_profits, na.rm = TRUE)
## [1] 17.5448

boxplot(
  movies$log_profits,
  main = "Distribution of Log Movie Profits",
```

```

xlab = "Movies",
ylab = "Natural logarithm of profit",
names = "Positive-profit movies",
col = "lightgreen"
)

```



Your Answer:

After replacing the log values for zero and negative profits with NA, the mean log profit is about 17.5448. We used `na.rm = TRUE` to leave out the missing values, so this mean only includes the 486 movies with positive profits.

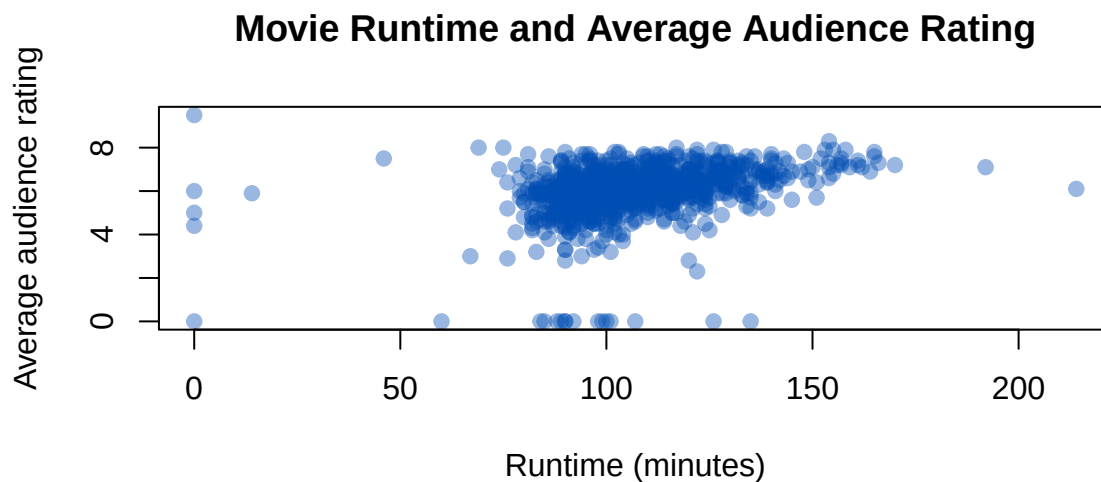
The new boxplot is less stretched towards high values because the logarithm brings large profits closer together. There are still several outliers on the lower side. The plots also include different movies: the first includes all movies, while the second only includes movies that made a positive profit.

Step f

```

# Create a scatterplot of runtime and average rating
plot(
  movies$runtime,
  movies$vote_average,
  main = "Movie Runtime and Average Audience Rating",
  xlab = "Runtime (minutes)",
  ylab = "Average audience rating",
  pch = 19,
  col = rgb(0, 0.3, 0.7, 0.4)
)

```



```
cor(movies$runtime, movies$vote_average)
## [1] 0.3204062
```

Your Answer:

The scatterplot suggests that longer movies tend to receive slightly higher ratings. The correlation is about 0.32, but the points are spread out, so there are plenty of exceptions.

Longer movies might have more time to develop their characters and stories, which could help explain the higher ratings. Genre and movie quality could also play a role. The plot does not prove that making a movie longer will improve its rating. There are also some zero values for runtime and rating that may represent missing or incorrectly recorded data. These values should be investigated before drawing firm conclusions.

Week 2

1 Population and sample

Is your dataset movies5.tsv the full population, or is it a sample of a larger population? If the latter, how would you describe the full population? [4 points]

Your Answer:

The dataset is a sample of a larger population because it contains 909 movies, rather than every movie. The full population would be all movies we want to draw conclusions about. For example, if we want to study movies released between 1990 and 2016, the population would include all movies released during those years. We cannot assume that the movies in our dataset were selected randomly.

2 Actors and movie revenue

Step a

For which actor in your data set do you observe the most movies? [2 points]

```
# Count the movies for each actor
actor_counts <- table(movies$first_actor)

head(sort(actor_counts, decreasing = TRUE))
##
##   Bruce Willis      Tom Hanks   Johnny Depp Sandra Bullock   Tom Cruise
##           8           8           7           7           7
## Drew Barrymore
##           6
```

Your Answer:

Bruce Willis and Tom Hanks appear most often, with 8 movies each. We use Bruce Willis for the next steps. These counts are based on the first_actor column, so they do not include appearances by actors elsewhere in a movie's cast.

Step b

What is the average revenue of the movie in which this actor plays and does the revenue lie above or below the revenue of an average movie according to your data set? [2 points]

```
# Average revenue of movies with Bruce Willis as first_actor
mean(movies$revenue[movies$first_actor == "Bruce Willis"],
     na.rm = TRUE)
## [1] 228062987

mean(movies$revenue, na.rm = TRUE)
## [1] 88846315
```

Your Answer:

The average revenue of the movies with Bruce Willis is 228,062,987.38. The average revenue across all movies in the dataset is 88,846,314.90. His movies therefore earn more revenue on average than the average movie in this dataset.

Step c

How trustworthy do you consider your conclusion to answer 2b? Use the term “law of large numbers” in your explanation. [2 points]

```
# Number of movies used for the Bruce Willis average
sum(movies$first_actor == "Bruce Willis", na.rm = TRUE)
## [1] 8
```

Your Answer:

The comparison is correct for this dataset, but it is based on only 8 Bruce Willis movies. One or two very successful movies could have a large effect on his average.

The law of large numbers tells us that, with a sufficiently large random sample, the sample mean tends to get closer to the population mean. Since we only have 8 movies and do not know whether they were randomly selected, we should

be careful about applying this result to all Bruce Willis movies. It also does not show that Bruce Willis himself causes higher revenue.

3 Population and sample variance

For this question, you will assume that your data set is the full population.

Step a

Recode profits such that it is expressed in millions. What is the variance of the variable profits (in millions) in your data set? [2 points]

```
# Calculate profits in millions
movies$profits <- (movies$revenue - movies$budget) / 1000000

# Calculate the population variance
population_variance <- mean(
  (movies$profits - mean(movies$profits))^2
)

population_variance
## [1] 17802.62
```

Your Answer:

The population variance of profits is approximately 17,802.62 squared million dollars. The variance is calculated using squared deviations from the mean, so negative and positive deviations do not cancel each other out. We divide by N, rather than N-1, because the assignment asks us to treat the dataset as the full population.

Step b

Create a new data set, called movies_sample. Make sure that it is a random sample of your data set of 25 movies. What is the variance of profits in this random sample? How does it compare to the variance of profits in 2a? [2 points]

```
# Make the random sample reproducible
set.seed(47)

movies_sample <- movies[sample(nrow(movies), 25), ]

var(movies_sample$profits)
## [1] 15728.93
```

Your Answer:

The sample variance is approximately 15,728.93 squared millions. This is lower than the population variance of 17,802.62 from the previous step.

The values differ because the sample contains only 25 of the 909 movies. A different sample could give a higher or lower variance, depending on which movies are selected.

Step c

In a for loop, create 100 different samples of 25 movies, as in b, and estimate the variance within each sample. Save the variance of each sample in a vector called sample_vars. So the first position of the vector would have the variance of the first sample, the second position the variance of the second sample, etc. Print the start of this vector. [2 points]

```
# Make the random samples reproducible
set.seed(47)

sample_vars <- numeric(100)

# Take 100 samples and save their variances
for (i in 1:100) {
  movies_sample <- movies[sample(nrow(movies), 25), ]
  sample_vars[i] <- var(movies_sample$profits)
}
```

```
head(sample_vars)
## [1] 15728.93 31971.87 16532.82 28257.89 24848.85 56289.00
```

Your Answer:

We took 100 random samples of 25 movies and saved each sample variance in `sample_vars`. The first six variances are approximately 15,728.93, 31,971.87, 16,532.82, 28,257.89, 24,848.85, and 56,289.00.

The variances differ because each sample contains a different selection of movies, as explained in step 3b.

Step d

Summarize and make a histogram of `sample_vars`. What is the mean, standard deviation and shape of its distribution? [2 points]

```
# Summarize the sample variances
summary(sample_vars)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   875.3   6303.8 16465.8 19982.6 26872.9 80456.9
```

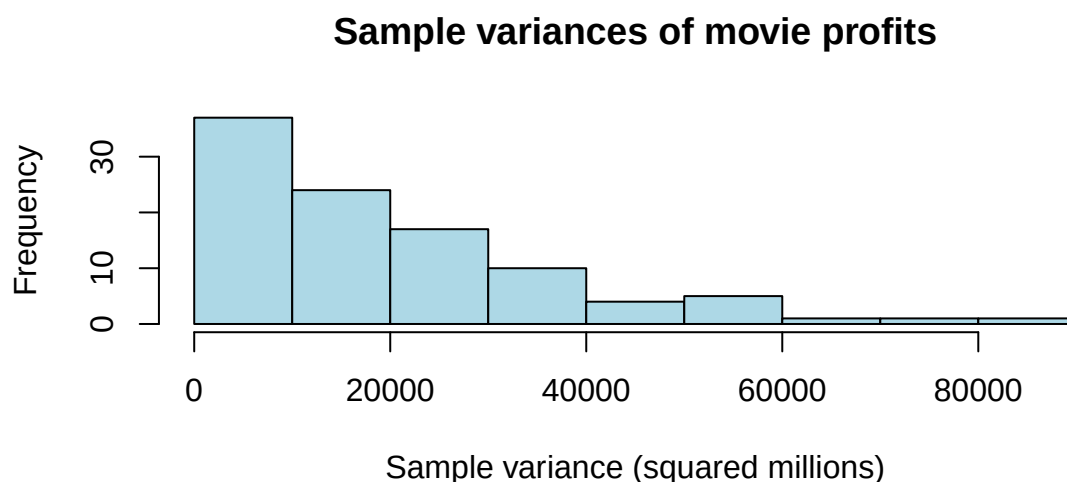
```
mean(sample_vars)
```

```
## [1] 19982.63
```

```
sd(sample_vars)
```

```
## [1] 16853.97
```

```
hist(sample_vars,
      main = "Sample variances of movie profits",
      xlab = "Sample variance (squared millions)",
      ylab = "Frequency",
      col = "lightblue")
```



Your Answer:

The mean of the 100 sample variances is approximately 19,982.63, and the standard deviation is approximately 16,853.97. Both are expressed in squared millions.

The histogram shows many lower sample variances and a smaller number of much higher values, creating a long tail to the right. Samples that include movies with exceptionally high profits can have much larger variances than samples without those movies, because their profits are far above the sample mean. These large differences are squared, giving them a strong effect on the result.

Step e

In your opinion, is a sample of 25 movies sufficient to get a reliable estimate of the population variance of profits, using the sample variance? Explain. [2 points]

Your Answer:

A sample of 25 movies does not seem large enough to give a reliable estimate of the population variance in this dataset. The sample variances differ a lot, and their standard deviation is large compared with their mean.

A few movies have exceptionally high profits. Including or missing one of these movies can strongly change the sample variance. A larger random sample would generally give a more stable estimate of the population variance, in this dataset.

Week 3

For the next part of the assignment, assume that the movies in your data frame are a random sample of a larger population of movies.

1 Confidence intervals for runtime

Step a

Create a new data set that only includes movies that are of the genre “Thriller”. For these thriller movies, give a 99 percent confidence interval for the variable runtime. Interpret the result. [2 points]

```
# Create a dataset with only thriller movies
movies_thriller <- subset(movies, genre == "Thriller")

n_thriller <- nrow(movies_thriller)
mean_runtime <- mean(movies_thriller$runtime)
n_thriller
## [1] 127
mean_runtime
## [1] 106.1969

# Calculate the 99% confidence interval for the mean runtime
test_runtime <- t.test(movies_thriller$runtime, conf.level = 0.99)
test_runtime$conf.int
## [1] 102.6930 109.7007
## attr(,"conf.level")
## [1] 0.99
```

Your Answer:

There are 127 thriller movies in total, with an average runtime of approximately 106.20 minutes. The 99% confidence interval for the population mean runtime is from 102.69 to 109.70 minutes.

We are 99% confident that the mean runtime of thriller movies in the population lies within this interval. This interval is about the population mean, rather than the runtime of an individual movie. We use a t-distribution because the population variance is unknown in this question.

Step b

Now, assume that the variance of runtime amongst thriller movies in your data is exactly the same as the variance of runtime in the population. Under this assumption, give a 99 percent confidence interval for the variable runtime among thriller movies. Interpret the result. Is your confidence interval wider or less wide than the one you found under question 1a? Why is that the case? [2 points]

```
# Use the sample variance as the known population variance
population_variance_runtime <- var(movies_thriller$runtime)

standard_error_runtime <- sqrt(population_variance_runtime / n_thriller)

# Critical z-value for a 99% confidence interval
z_value <- 2.576

lower_runtime <- mean_runtime - z_value * standard_error_runtime
upper_runtime <- mean_runtime + z_value * standard_error_runtime

c(lower_runtime, upper_runtime)
## [1] 102.7458 109.6479
```

Your Answer:

Under this assumption, the 99% confidence interval is from 102.75 to 109.65 minutes. This means that we are 99% confident that the true mean runtime of all thriller movies lies between those two values.

This interval is a little bit narrower than the interval in step a. In this question, we assume that the population variance is known, so we use the normal distribution instead of the t-distribution as we did earlier. For a 99% confidence interval, we get a critical z-value of 2.576. Because this value is slightly smaller than the t-value used in the previous step, the confidence interval is also slightly narrower.

2 Testing the variance of runtime

Step a

Using an appropriate five-step procedure, set up a test for the null hypothesis that the variance of runtime equals 500. Clearly state your null hypothesis, alternative hypothesis, your test statistic, your critical value, and your conclusion. [2 points]

```
# Number of movies
n <- nrow(movies)

sample_variance <- var(movies$runtime)

# Calculate the chi-square test statistic
chi_square <- (n - 1) * sample_variance / 500

# Critical values for a 5% two-sided test
critical_values <- qchisq(c(0.025, 0.975), df = n - 1)

sample_variance
## [1] 396.6046
chi_square
## [1] 720.234
critical_values
## [1] 826.3872 993.4009
```

Your Answer:

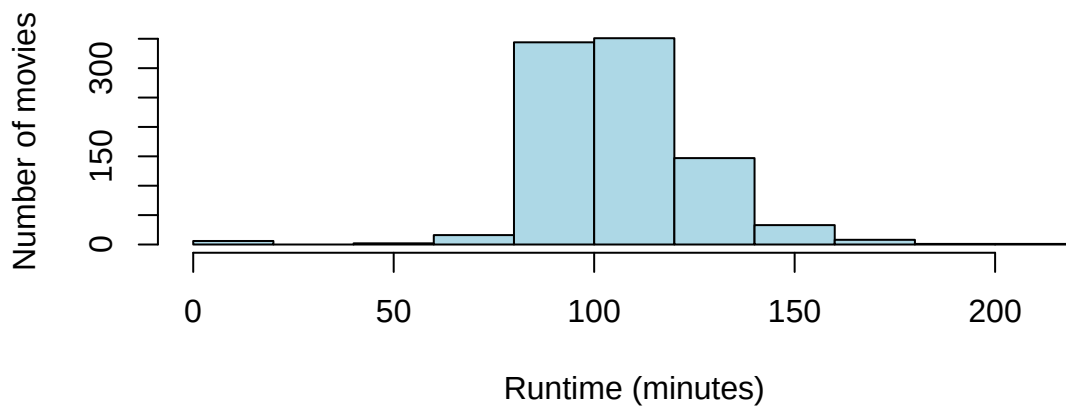
1. The null hypothesis is H_0 : the population variance of runtime equals 500 minutes squared. The alternative hypothesis is H_1 : the population variance is not equal to 500 minutes squared.
2. We use a significance level of 5% and a two-sided chi-squared test. The test assumes that runtime is normally distributed in the population.
3. There are 909 movies in the sample. The sample variance is approximately 396.60 minutes squared. The test statistic is calculated as $(909 - 1)$ times the sample variance divided by 500, which gives 720.23. Under H_0 and the normality assumption, this statistic follows a chi-squared distribution with 908 degrees of freedom.
4. The critical values are 826.39 and 993.40. We reject H_0 if the test statistic is below 826.39 or above 993.40. Otherwise, we do not reject H_0 .
5. Since 720.23 is below 826.39, we reject H_0 at the 5% significance level, assuming that runtime is normally distributed. The sample variance of 396.60 is lower than the hypothesized population variance of 500 minutes squared. In step b, we check whether the normality assumption seems reasonable.

Step b

For the validity of your test in 2a, what assumption about the distribution of runtime needs to hold? Make an appropriate plot to test this assumption. What do you conclude? [2 points]

```
# Create a histogram to check whether runtime is approximately normal
hist(movies$runtime,
     main = "Distribution of movie runtime",
     xlab = "Runtime (minutes)",
     ylab = "Number of movies",
     col = "lightblue")
```

Distribution of movie runtime



Your Answer:

The chi-squared test in step a relies on the idea that runtime follows a normal distribution in the population. To check this, we looked at a histogram. There's a clear cluster of movies between 80 and 120 minutes, and the center does look somewhat bell-shaped. But the right side stretches out much further, and there are a few very short runtimes—some are even zero. So, the distribution isn't clearly normal. With this in mind, the normality assumption is shaky, and we need to be cautious about any conclusions from step a.

3 Movie ratings and profits

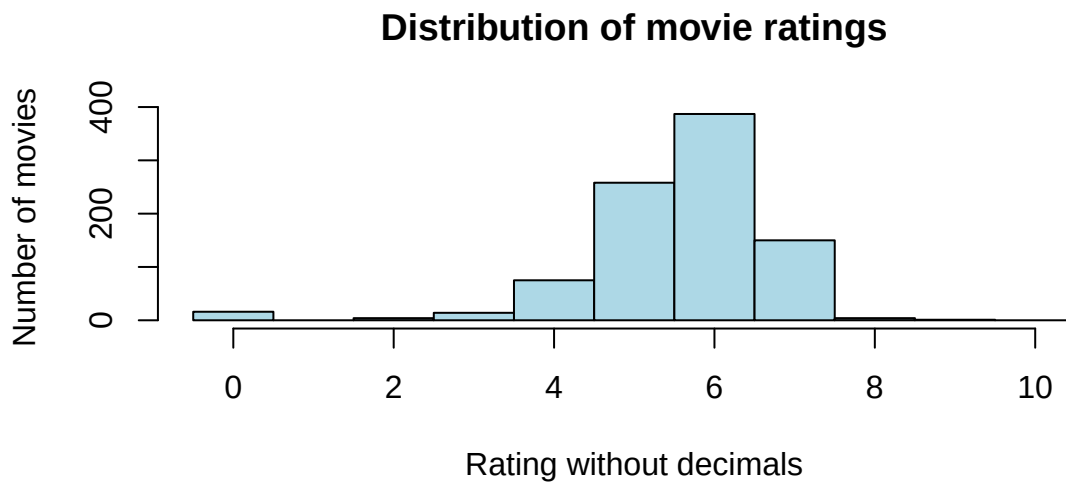
There is an argument going on in the movie studio. Bob claims that they should make higher-quality movies, as this will bring in more profits. Chantal disagrees. She tells Bob that mediocre movies bring the most profits. You are asked to advise on who is right.

Step a

Create a new variable called `vote_average_rounded`. Make sure this variable is the same as `vote_average`, but without any decimals (i.e., a 6.3 becomes a 6, a 8.7 an 8, etc.). Display a histogram of `vote_average_rounded`. [2 points]

```
# Remove decimals, following the examples in the question
movies$vote_average_rounded <- floor(movies$vote_average)

hist(movies$vote_average_rounded,
     breaks = seq(-0.5, 10.5, by = 1),
     main = "Distribution of movie ratings",
     xlab = "Rating without decimals",
     ylab = "Number of movies",
     col = "lightblue")
```



```
table(movies$vote_average_rounded)
##
##  0  2  3  4  5  6  7  8  9
## 16  4 14 75 258 387 150  4  1
```

Your Answer:

We removed the decimals using `floor()`, so 6.3 becomes 6 and 8.7 becomes 8, as in the question. Most movies are in rating groups 5, 6 and 7. Group 6 is the largest, with 387 movies. There are only 4 movies in group 8 and 1 movie in group 9.

Step b

Create a scatter plot with `vote_average_rounded` on the x axis and the mean of profits within each category of `vote_average_rounded` on the y-axis. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. At which rating of movies are the profits the highest? [3 points]

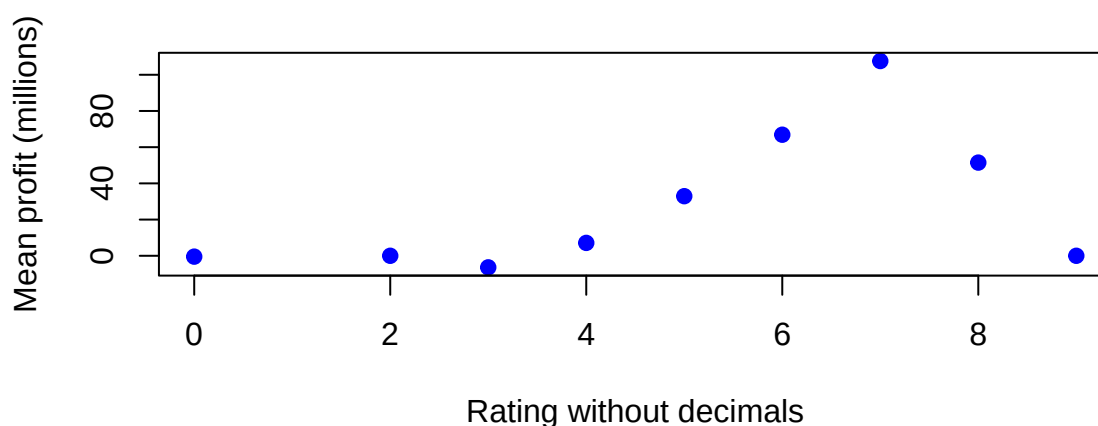
```
# Calculate the mean profit for each rating group
# Profits are already expressed in millions in week 2
profits_by_rating <- aggregate(profits ~ vote_average_rounded,
                               data = movies, FUN = mean)

profits_by_rating
##  vote_average_rounded  profits
## 1                    0 -0.4679164
```

```
## 2      2  0.000000
## 3      3 -6.390935
## 4      4  7.117685
## 5      5 32.931806
## 6      6 66.888337
## 7      7 107.585056
## 8      8 51.482190
## 9      9  0.000000

plot(profits_by_rating$vote_average_rounded,
     profits_by_rating$profits,
     main = "Average movie profits by rating",
     xlab = "Rating without decimals",
     ylab = "Mean profit (millions)",
     pch = 19,
     col = "blue")
```

Average movie profits by rating



```
profits_by_rating[which.max(profits_by_rating$profits), ]
##  vote_average_rounded  profits
## 7                    7 107.5851
```

Your Answer:

Rating group 7 has the highest mean profit, at approximately 107.59 million. This group contains movies with original ratings from 7.0 up to (but not including) 8.0.

Mean profits generally increase as you move from the lower rating groups to group 7. After that, profits drop off for groups 8 and 9. So, in this given sample, the movies with the highest ratings don't actually bring in the highest average (mean) profits.

Step c

Recreate the scatter plot with year on the x axis and mean profits on the y-axis, but now add bars around each point, indicating the 95% confidence interval. [3 points]

```
# List the release years from lowest to highest
years <- sort(unique(movies$release_year))

mean_profits_year <- numeric(length(years))
lower_profits_year <- numeric(length(years))
upper_profits_year <- numeric(length(years))

# Calculate the mean profit and 95% confidence interval for each year
for (i in 1:length(years)) {
  year_profits <- movies$profits[movies$release_year == years[i]]

  mean_profits_year[i] <- mean(year_profits)
```

```

standard_error_year <- sd(year_profits) / sqrt(length(year_profits))

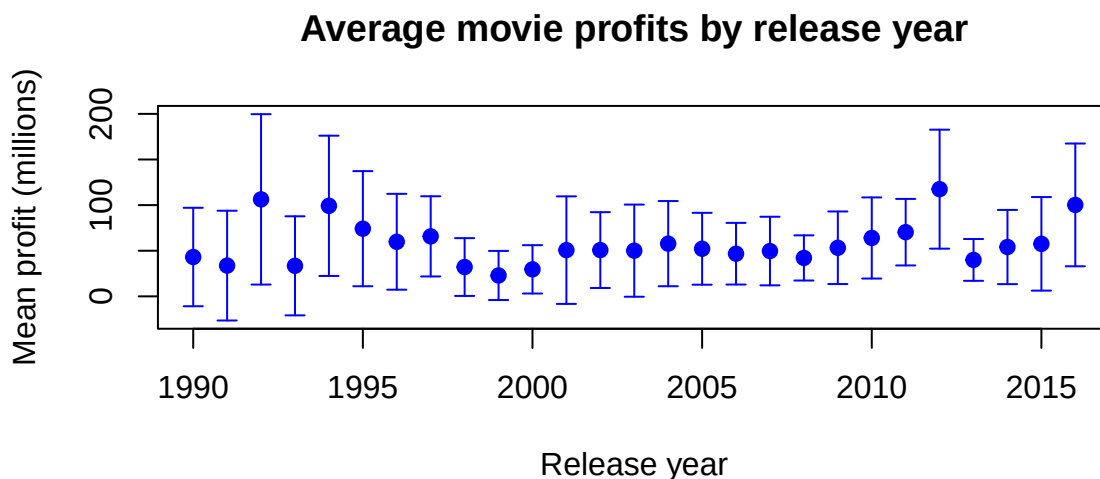
margin_year <- qt(0.975, df = length(year_profits) - 1) *
  standard_error_year

lower_profits_year[i] <- mean_profits_year[i] - margin_year
upper_profits_year[i] <- mean_profits_year[i] + margin_year
}

plot(years, mean_profits_year,
     main = "Average movie profits by release year",
     xlab = "Release year",
     ylab = "Mean profit (millions)",
     ylim = range(lower_profits_year, upper_profits_year),
     pch = 19,
     col = "blue")

# Add the confidence interval bars
arrows(x0 = years, y0 = lower_profits_year,
       x1 = years, y1 = upper_profits_year,
       angle = 90, code = 3, length = 0.05,
       col = "blue")

```



Your Answer:

The plot shows the mean profit for each release year, with bars showing its 95% confidence interval. The highest mean profit is in 2012, at approximately 117.49 million. Its confidence interval runs from 52.28 to 182.70 million.

Many intervals are wide, showing that the estimates are uncertain. Movie profits vary a lot, and some years contain few movies. These t-intervals are approximate because profits are strongly skewed, especially for years with small samples. This plot compares release years, as requested in the question.

Step d

Write an advice to settle the argument between Bob and Chantal. [4 points]

Your Answer:

Our results do not clearly show that either Bob or Chantal is right. Rating group 7 has the highest average profit, at about 107.59 million. Groups 5 and 6 have lower average profits of 32.93 million and 66.89 million. If we consider groups 5 and 6 to be mediocre, this does not support Chantal's claim in this sample.

However, higher ratings do not always go together with higher profits. Group 8 has an average profit of 51.48 million. Group 9 has a calculated average profit of zero, based on the recorded revenue and budget. Both values are zero for its only movie, so we cannot conclude that it actually earned nothing. Groups 8 and 9 contain only 4 movies and 1 movie, so we cannot draw strong conclusions about these groups.

We advise the studio to make movies that audiences enjoy, but a higher rating does not guarantee more profit. Budget, genre and marketing may also matter. These results do not prove that better movies cause higher profits. The confidence intervals in step c compare release years, so they do not show whether differences between rating groups are statistically significant.